

Definition 6: A function f whose domain and codomain are subsets of the set of real numbers is called increasing if $f(x) < f(y)$, and strictly increasing if $f(x) < f(y)$, whenever $x < y$ and x and y are in the domain of f . Similarly, f is called decreasing if $f(x) > f(y)$, and strictly decreasing if $f(x) > f(y)$, whenever $x < y$ and x and y are in the domain of f . (The word strictly in this definition indicates a strict inequality.)

Example 12: The function $f(x) = x^2$ from $\mathbb{R}^+$ to $\mathbb{R}^+$ is strictly increasing.

Let, $x, y \in \mathbb{R}^+$ and WLOG, $x < y$. $\therefore \exists z \in \mathbb{R}^+$ s.t. $y = x + z$.

$$f(y) = y^2 = (x+z)^2 = x^2 + 2xz + z^2 = f(x) + 2xz + z^2 = f(x) + z(2x+z).$$

It's easy to see that $z(2x+z) > 0$. $\therefore f(y) > f(x)$. Since, x and y are arbitrarily chosen, $f(x) = x^2$ from $\mathbb{R}^+$ to $\mathbb{R}^+$ is strictly increasing. $\square$

From these definitions, it can be shown that a function that is either strictly increasing or decreasing, strictly must be one-to-one. However, a function that is increasing, but not strictly increasing, or decreasing, but not strictly decreasing, is not one-to-one.

Definition 7: A function f from A to B is called onto, or a surjection, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called surjective if it is onto.

Remark: A function f is onto if $\forall y \exists x (f(x) = y)$, where the domain for x is the domain of the function and the domain for y is the codomain of the function.

Example 13: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3, f(b) = 2, f(c) = 1$ and $f(d) = 3$. Is f an onto function?

Solution: Because all three elements of the codomain are images of the elements in the domain, we see that f is onto. Note that if the codomain were $\{1, 2, 3, 4\}$, then f would not be onto. $\square$

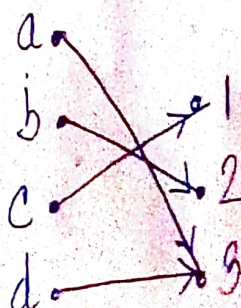


Figure 4: An onto function